

# Community Skills in Indian Knowledge System

A syllabus-based bilingual handbook for identifying traditional community knowledge, documenting processes and tools, proposing practical improvements, and applying socially relevant technologies in water, waste, energy, agriculture and irrigation.

COURSE CODE

4001

SEMESTER

4

CREDITS

No Credit

CATEGORY

Common Course

PERIODS / WEEK

2 (L:2, T:0, P:0)

PERIODS / SEMESTER

30

ASSESSMENT

CIA / Record / Fieldwork

COGNITIVE LEVEL

Applying

## 30 Hours of community learning

Observe respectfully, document accurately, improve appropriately and return useful knowledge to the community.

### COURSE PURPOSE

## What this course expects from students

### Appreciate living knowledge

Recognise the skills practised by artisans, workers, farmers and rural communities near the institution.

സമാപനത്തിന്റെ സമീപപ്രദേശങ്ങളിലെ കരകൗശലവിദഗ്ധർ, കർഷകർ, തൊഴിലാളികൾ, ഗ്രാമസമൂഹങ്ങൾ എന്നിവരുടെ പാരമ്പര്യജ്ഞാനവും കഴിവുകളും ആദരവോടെ തിരിച്ചറിയുക.

## Connect tradition and technology

Relate traditional processes, tools and implements to the technical knowledge gained in the diploma programme.

Traditional process, tools, implements എന്നിവയെ diploma-യിൽ പഠിച്ച technical knowledge-ുമായി ബന്ധിപ്പിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കാം.

## Propose useful improvements

Suggest changes that improve safety, quality, productivity, income, comfort and sustainability without destroying traditional value.

പാരമ്പര്യരീതിയുടെ മൂല്യം നഷ്ടപ്പെടുത്താതെ safety, quality, productivity, income, comfort, sustainability എന്നിവ മെച്ചപ്പെടുത്തുന്ന നിർദ്ദേശങ്ങൾ നൽകുക.

## Use socially relevant technologies

Practise or popularise low-cost scientific methods for water, waste and energy management, agriculture and irrigation.

Water, waste, energy management, agriculture, irrigation മേഖലകളിൽ community-ക്ക് പ്രയോജനമുള്ള low-cost scientific methods പ്രയോഗിക്കുക അല്ലെങ്കിൽ പ്രചരിപ്പിക്കുക.

## Document ethically

Obtain consent before photographs, video or personal details and avoid exposing confidential information.

Photo/video എടുക്കുന്നതിന് മുമ്പ് permission വാങ്ങണം. വ്യക്തിഗത വിവരങ്ങളും രഹസ്യമായ തൊഴിൽരീതികളും അനുമതിയില്ലാതെ പ്രസിദ്ധീകരിക്കരുത്.

## Return value

Share useful findings, suggestions or awareness material with the participating community whenever possible.

Study പൂർത്തിയാക്കിയ ശേഷം findings, suggestions, awareness material എന്നിവ community-യുമായി പങ്കിടുന്നത് നല്ല practice ആണ്.

# Course-outcome and module map

The four modules total 30 hours and lead from observation to documentation, improvement and community application.

|                  |          |
|------------------|----------|
| CO1 / Module I   | 10 hours |
| CO2 / Module II  | 6 hours  |
| CO3 / Module III | 10 hours |
| CO4 / Module IV  | 4 hours  |

| CO  | Outcome                                                                                              | Duration | Level    |
|-----|------------------------------------------------------------------------------------------------------|----------|----------|
| CO1 | Identify knowledge, skills and practices followed traditionally.                                     | 10 hours | Applying |
| CO2 | Explain processes, methods and implements followed traditionally.                                    | 6 hours  | Applying |
| CO3 | Identify improvements in processes and tools to enhance productivity and community living standards. | 10 hours | Applying |
| CO4 | Use socially relevant technologies in water, waste and energy management for the community.          | 4 hours  | Applying |

**Prerequisites:** Art, culture and traditions from secondary-school Social Science, plus programme-relevant technical knowledge acquired up to Semester 3. CO1 and CO2 are strongly mapped to PO5 and PO7, CO3 to PO5, and CO4 to PO5 and PO7.

**Editorial clarification:** Module IV on page 3 repeats “water” in one sentence. The objectives and CO4 consistently identify the intended areas as **water, waste and energy management**; this lesson follows that interpretation.

CO1 • 10 HOURS

## Module I — Identify Traditional Knowledge, Skills and Practices

Conduct a structured local field visit to recognise what the community knows, how the skill is learned, and why the practice remains useful.

Step 1

Choose responsibly

### Select a suitable local practice

Choose a skill genuinely practised in the locality, observable safely, and supported by a willing knowledge holder. Examples include pottery, coir work, handloom, bamboo or cane craft, metalwork, carpentry, traditional food processing, indigenous building, farming, seed preservation, irrigation, water harvesting and repair practices.

**മലയാളം:** Locality-ൽ യഥാർത്ഥമായി ചെയ്യുന്ന skill തിരഞ്ഞെടുക്കുക. കാണാൻ കഴിയണം, സുരക്ഷിതമായിരിക്കണം, practitioner പഠനത്തിന് സമ്മതിക്കണം.

Step 2

Plan before visiting

Prepare the visit

- ✓ Obtain faculty approval and identify the team.
- ✓ Fix date, time, place and contact person.
- ✓ Prepare questionnaire and observation sheet.
- ✓ Carry notebook, pen, measuring tape and permitted camera.
- ✓ Identify safety needs such as footwear, mask or gloves.
- ✓ Prepare a simple consent statement.

**മലയാളം:** Visit-ിന് മുമ്പ് date, place, contact person, questionnaire, safety, photo permission എന്നിവ തയ്യാറാക്കണം.

Step 3

Observe systematically

What to observe

| Area                | Questions                                                          | Evidence                                             |
|---------------------|--------------------------------------------------------------------|------------------------------------------------------|
| Knowledge source    | How was the skill learned—family, apprenticeship or community?     | Interview note and learning timeline.                |
| Raw materials       | What is used, where is it sourced, and is it seasonal?             | Material list and approximate quantity.              |
| Tools               | What is each tool called and what does it do?                      | Labelled sketch or permitted photograph.             |
| Process             | What are the steps, sequence, time and quality checks?             | Flowchart and time observations.                     |
| Skill judgement     | Which steps rely on touch, sound, colour or experience?            | Practitioner explanation and student interpretation. |
| Safety              | Are there heat, dust, sharp tools, heavy loads or awkward posture? | Hazard list without blaming the worker.              |
| Economics           | What affects cost, demand, output and income?                      | Only voluntarily shared approximate data.            |
| Culture/environment | What local identity or ecological value does the practice carry?   | Context note and examples.                           |

**മലയാളം:** Materials, tools, process sequence, skill judgement, safety, cost, cultural value, environmental impact എന്നിവ വേർതിരിച്ച് രേഖപ്പെടുത്തുക.

1. Name of the skill/practice:
2. Name of artisan/community group (only with consent):
3. Location and date of visit:
4. How long has this practice existed in the family/community?
5. How did the practitioner learn the skill?
6. What raw materials are used and where are they sourced?
7. What tools and implements are used? State each function.
8. What are the main process steps?
9. Which step requires the greatest experience or judgement?
10. What quality checks are followed?
11. What difficulties are faced—time, cost, safety, demand, material or transport?
12. Have tools or methods changed in recent years?
13. What improvement would the practitioner welcome?
14. What should not be changed because it is essential to quality or tradition?
15. May photographs/video be used in the academic record? Yes / No / Limited.

**CO1 evidence:** approved field-visit plan, completed questionnaire, observation notes, labelled evidence and identification of the traditional knowledge embedded in the practice.

CO2 • 6 HOURS

## Module II — Explain the Process, Methods and Implements

Convert raw field observations into a clear report supported by appropriate photographs, sketches, tables and videos.

Report Structure

6 hours • Applying

### Recommended report order

- 1 **Cover page:** course, title, student details, institution and academic year.
- 2 **Certificate:** bonafide statement signed as required.
- 3 **Acknowledgement:** thank practitioner, community, guide and institution.
- 4 **Abstract:** 150–250 words covering skill, method, finding and improvement.
- 5 **Introduction:** locality, background, relevance and objectives.
- 6 **Methodology:** visit, observation, interview, questionnaire and documentation.
- 7 **Traditional process:** materials, tools, steps, skills and quality checks.

**8 Findings:** strengths, limitations, safety, productivity and environment.

**9 Suggestions:** technically justified, low-cost and acceptable improvements.

**10 Conclusion and annexures:** learning, questionnaire, diary, evidence and consent.

**മലയാളം:** Report ഒരു ക്രമപോലെ അല്ല; Introduction → Methodology → Process → Findings → Improvements → Conclusion എന്ന technical order പാലിക്കുക.

### Photo Evidence

## Use photographs correctly

- ✓ Use only permitted images.
- ✓ Show process, tool or result—not unnecessary personal details.
- ✓ Add figure number, caption, location and date.
- ✓ Refer to every important figure in the text.
- ✓ Do not stretch, blur or over-edit evidence.

**Caption:** “Figure 3. Manual shaping stage using a locally fabricated wooden tool.”

Photo-ക്ക് figure number, caption, context നൽകണം. Permission ഇല്ലാത്ത face/photo ചേർക്കരുത്.

### Video

## Use video in presentation

- ✓ Keep clips short and relevant.
- ✓ Introduce the process before playing.
- ✓ Remove private conversation.
- ✓ Add subtitles where sound is unclear.
- ✓ Explain the technical point shown.

Video മാത്രം കാണിക്കുന്നത് explanation-ന് പകരമല്ല. “ഇത് എന്താണ് തെളിയിക്കുന്നത്?” എന്ന് പറയണം.

### Process Map

## Convert observation into an engineering record

For each step, record input, tool, action, approximate time, skill judgement, quality check, output and hazard.

മലയാളം: complete process step-നും input, tool, action, time, skill, quality check, output, hazard എന്നിവ എഴുതുക.

**CO2 evidence:** complete process flow, tool table, labelled documentary evidence, clear report writing and a short presentation.

CO3 • 10 HOURS

## Module III — Improve Processes and Tools

Identify improvements that are technically sound, affordable, safe, respectful and acceptable to the community.

Improvement Cycle

10 hours • Applying

### From observation to recommendation

- 1 Define the problem:** state what happens, where, how often and who is affected.
- 2 Protect traditional value:** identify what must not be lost—quality, identity, skill or cultural meaning.
- 3 Find root causes:** check material, method, tool, person, environment and measurement.
- 4 Generate options:** produce more than one solution, including a low-cost option.
- 5 Evaluate feasibility:** cost, safety, maintenance, energy, training, local availability, acceptance and benefit.
- 6 Prototype:** test a small change before recommending full adoption.
- 7 Measure impact:** compare time, output, defects, effort, safety or resource use.

**മലയാളം:** Problem കണ്ട ഉടൻ expensive machine suggest ചെയ്യരുത്. Root cause കണ്ടെത്തി low-cost, maintainable, locally acceptable option തിരഞ്ഞെടുക്കുക.

Improvement Areas

# What can be improved?

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| Area         | Possible improvement                                                                       | Evidence                                    |
|--------------|--------------------------------------------------------------------------------------------|---------------------------------------------|
| Safety       | Guard, stable surface, ventilation, dust control, lighting, safer handle or PPE awareness. | Hazard observation and risk reduction.      |
| Ergonomics   | Correct work height, seating, tool grip and reduced bending.                               | Posture comparison and user feedback.       |
| Productivity | Layout, jig, fixture, template, batch organisation or reduced waiting.                     | Time study or output comparison.            |
| Quality      | Gauge, size guide, moisture check, temperature indicator or checklist.                     | Defect/rework comparison.                   |
| Material use | Accurate cutting, off-cut reuse, storage or controlled mixing.                             | Waste estimate or material balance.         |
| Energy       | Insulation, improved airflow, efficient motor, solar drying or daylight use.               | Fuel/electricity estimate and safety check. |
| Marketing    | Labelling, packaging, digital catalogue and direct customer link.                          | Community interest and practical plan.      |

## Feasibility

### Score each proposal

**1–5**

Cost

**1–5**

Safety

**1–5**

Benefit

**1–5**

Maintenance

**1–5**

Acceptance

Explain each score and select the option with the best practical balance—not merely the highest technology.

Technology advanced ആണെന്ന് മാത്രം criterion അല്ല. Cost, benefit, maintenance, acceptance എല്ലാം നോക്കണം.

## Recommendation

### Write recommendations correctly

- ✓ State the observed issue.
- ✓ Give the technical reason.
- ✓ Describe the proposed modification.



- ✓ Estimate materials, cost and maintenance.  
Course 4001
- ✓ State expected measurable benefit.
- ✓ Mention limitations and training.
- ✓ Record practitioner response.

“Improve ചെയ്യണം” എന്ന് മാത്രം എഴുതരുത്. എന്ത് change, എങ്ങനെ, എത്ര cost, എന്ത് benefit എന്ന് വ്യക്തമാക്കണം.

**CO3 evidence:** detailed case study, present-condition analysis, at least two alternatives, feasibility comparison, final recommendation and expected impact.

CO4 • 4 HOURS

## Module IV — Socially Relevant Technology

Practise, popularise or improve scientific methods in water, waste and energy management, agriculture and irrigation, supported by documentary evidence.

### Water

## Water management

- ✓ Leak detection and repair awareness.
- ✓ Rainwater collection and first-flush awareness.
- ✓ Safe storage and covered containers.
- ✓ Simple water-use audit.
- ✓ Recharge and runoff control appropriate to the site.

Water project-ൽ local conditions, water quality, safety, maintenance എന്നിവ പരിശോധിക്കണം.

### Waste

## Waste management

- ✓ Source segregation into appropriate categories.
- ✓ Composting suitable biodegradable waste.
- ✓ Safe collection of sharp, chemical or electronic waste.
- ✓ Material reuse and reduction at source.
- ✓ Awareness signage and monitoring record.

Hazardous/e-waste സാധാരണ compost/reuse stream-ൽ ചേർക്കരുത്.

## Energy

### Non-conventional and efficient energy

- ✓ Solar drying or lighting demonstration.
- ✓ Improved daylight and ventilation.
- ✓ Efficient cooking or process heat.
- ✓ Energy-use survey and awareness.
- ✓ Biogas only with expert guidance.

Electrical/gas work expert supervision ഇല്ലാതെ ചെയ്യരുത്.

## Agriculture

### Agriculture and irrigation

- ✓ Mulching and moisture conservation.
- ✓ Drip or controlled irrigation demonstration.
- ✓ Simple soil-moisture observation.
- ✓ Seed and produce storage improvement.
- ✓ Tool ergonomics and safe handling.

Farmer experience ആദരിക്കുക; suggestion field condition-നോട് പൊരുത്തപ്പെടുന്നുണ്ടോ verify ചെയ്യണം.

## Project Plan

### Four-hour community activity format

- 1 Need:** select one real and manageable issue with the community.
- 2 Baseline:** record present condition using count, measurement, photograph or survey.
- 3 Intervention:** demonstrate, install, practise or popularise the selected method safely.
- 4 Awareness:** provide a bilingual poster, checklist or demonstration.
- 5 Verification:** record participation, feedback and measurable result.

**Documentation:** include before/after evidence, limitations and follow-up.

Activity ചെറിയതായാലും real community benefit കാണിക്കണം. Before/after evidence, feedback, limitations രേഖപ്പെടുത്തുക.

**CO4 evidence:** need statement, baseline, safe intervention, awareness material, documentary evidence, feedback and follow-up recommendation.

## CIA RECORD HANDBOOK

# Complete record and submission structure

The syllabus states that CIA is to be followed as included in the handbook. This section organises the evidence expected from a field-oriented course.

## Before fieldwork

- ✓ Topic approval
- ✓ Team and roles
- ✓ Visit plan
- ✓ Questionnaire
- ✓ Safety and consent plan

## During fieldwork

- ✓ Dated field diary
- ✓ Interview notes
- ✓ Process timing
- ✓ Tool sketches
- ✓ Permitted photo/video

## After fieldwork

- ✓ Process flowchart
- ✓ Analysis tables

## Bonafide report contents page

1. Cover Page
2. Bonafide Certificate
3. Student Declaration
4. Acknowledgement
5. Abstract
6. Table of Contents
7. List of Figures and Tables
8. Introduction and Objectives
9. Profile of Locality / Community / Traditional Skill
10. Methodology
11. Raw Materials, Tools and Implements
12. Detailed Process Description
13. Traditional Knowledge and Skill Elements
14. Findings: Quality, Safety, Productivity, Environment and Economics
15. Problems Identified
16. Alternative Improvements
17. Feasibility Analysis
18. Final Suggestions and Recommendations
19. Socially Relevant Technology Activity
20. Results / Community Feedback
21. Conclusion
22. References
23. Annexure A — Questionnaire
24. Annexure B — Field Diary
25. Annexure C — Consent / Photo Permission
26. Annexure D — Documentary Evidence

### Possible CIA criterion

### Quality evidence

|                         |                                                                                  |
|-------------------------|----------------------------------------------------------------------------------|
| Participation           | Prepared visit, respectful interaction, observation and individual contribution. |
| Technical understanding | Correct process, labelled tools, flowchart and programme connection.             |
| Documentation           | Clear record, captions, dates, questionnaire and ethical consent.                |
| Analysis                | Root-cause reasoning, alternatives, feasibility and measurable benefit.          |
| Community relevance     | Practical, safe, affordable and acceptable recommendation.                       |
| Presentation            | Clear explanation, team coordination and honest limitations.                     |

**Do not fabricate evidence:** A smaller genuine study with real dates, observations and limitations is better than a large report containing invented interviews, costs, photographs or results.

COMPLETE MODEL CASE STUDY

# Hypothetical sample — Traditional pottery process improvement

This model demonstrates report structure and reasoning. Students must replace every detail with verified field observations.

## TITLE

Study of a Traditional Pottery Process and Proposal for a Low-Cost Ergonomic and Drying Improvement

## ABSTRACT

This hypothetical study documents clay preparation, wedging, wheel forming, initial drying, finishing, firing and inspection. It focuses on knowledge used to judge clay consistency, control hand pressure and identify drying condition. Two possible problems are analysed: repeated bending during clay preparation and cracking caused by uneven drying. A raised work platform and shaded ventilated drying rack are proposed. The options are evaluated for cost, safety, maintenance, local material availability and acceptance. Actual student reports must use verified field data.

## OBJECTIVES

1. Identify traditional knowledge and tools used in pottery.
2. Explain the process and skill-dependent stages.
3. Identify safety, ergonomic and productivity issues.
4. Develop low-cost improvement proposals.
5. Prepare evidence and recommendations.

## METHODOLOGY

A real project should use an approved visit, practitioner interview, observation, time notes, tool sketches, permitted photographs and feedback discussion. Personal or commercial information must be recorded only with consent.

## PROCESS FLOW

Clay selection → stone removal → water addition → kneading/wedging → wheel forming → initial drying → finishing → final drying → firing → inspection.

## TRADITIONAL KNOWLEDGE

- Judging clay moisture by touch and behaviour.
- Coordinating wheel speed and hand pressure.
- Recognising drying condition from colour and surface feel.
- Arranging products during firing through experience.
- Repairing small defects before complete drying.

## TOOLS

Potter's wheel, wooden shaping tools, cutting wire, water container, finishing tool, drying surface and kiln/firing arrangement.

## PROBLEM 1 — ERGONOMIC STRAIN

Condition to verify: low-level clay preparation may require repeated bending.

Root causes: low work height, poor material placement and manual handling.

Alternative A: raised stable preparation platform using local material.

Alternative B: low-cost seating and repositioned material containers.

Recommendation: stable raised platform after confirming preferred height.

Expected benefit: reduced bending and easier handling.

Limitation: it must not obstruct the traditional method and must remain stable when wet.

## PROBLEM 2 — UNEVEN DRYING

Condition to verify: direct sun or uneven airflow may cause non-uniform drying and cracking.

Root causes: rapid moisture loss, wet-ground contact and insufficient protected space.

Alternative A: shaded ventilated rack with multiple levels.

Alternative B: movable light cover and regular turning schedule.

Recommendation: locally fabricated shaded rack with a drying checklist.

Expected benefit: more uniform drying, cleaner storage and reduced handling damage.

## FEASIBILITY

Raised platform — Cost 4/5, Safety 4/5, Benefit 4/5, Maintenance 5/5, Acceptance to be confirmed.

#### SOCIALLY RELEVANT TECHNOLOGY LINK

A rainwater collection container for non-potable process use may be studied only after checking roof cleanliness, first-flush arrangement, safe storage and local suitability.

#### COMMUNITY FEEDBACK

A real report must record the practitioner's response: useful / not useful / needs modification, preferred dimensions and material, cost concern and willingness for a small trial.

#### CONCLUSION

Traditional pottery contains significant material, process and quality knowledge. Engineering improvement should support the practitioner rather than replace valuable judgement. Suitable recommendations are low-cost, safe, repairable and accepted by the user.

#### EVIDENCE TO ATTACH

Field plan, questionnaire, tool sketch, process flowchart, permitted photographs, measurements, feasibility matrix, practitioner feedback and team reflection.

**മലയാളം:** Model report copy ചെയ്യരുത്. സ്വന്തം field visit data, actual photos, practitioner feedback, measurements എന്നിവ ഉപയോഗിക്കുക.

### PRACTICE VIVA / WRITTEN REVIEW

## Questions with model answers

This is a CIA and viva practice set; the syllabus identifies the subject as a fieldwork-oriented no-credit course.

### 1. What is traditional knowledge?

It is knowledge, skill, judgement and practice developed and transmitted within a community over time, including material selection, tool use, process sequence, quality judgement and cultural meaning.

Community-യിൽ തലമുറകളായി വികസിക്കുകയും കൈമാറുകയും ചെയ്ത അറിവ്, skill, judgement, practice എന്നിവയാണ് traditional knowledge.

### 2. Why is consent necessary?

Consent respects privacy, rights and ownership of the knowledge holder and clarifies whether names, photographs, videos and process details may be used academically.

### 3. Observation and interview evidence—what is the difference?

Observation is what the student directly sees, measures or records. Interview evidence is the practitioner's explanation, experience or opinion. A report should distinguish them.

4. What is appropriate technology?

A safe, affordable, locally maintainable, resource-conscious, culturally acceptable and useful improvement under actual working conditions.

5. How can productivity be measured?

By time per unit, units per day, waiting time, material loss, defects, rework, effort or energy use, depending on the process.

6. What belongs in a process flowchart?

Inputs, preparation, core processing, finishing, output, inspections, waiting stages, rework and relevant waste streams.

7. Give two socially relevant technology examples.

A water-use audit with leak-reduction awareness; source segregation and composting of suitable biodegradable waste; solar drying; controlled-irrigation demonstration.

8. Why not replace traditional methods immediately?

A method may contain hidden quality, cultural, environmental or market value. Replacement without participation can reduce identity or create maintenance and cost problems.

9. What documentary evidence is expected?

Dated notes, questionnaire, sketches, permitted photographs/video, measurements, process flow, consent, calculations, feasibility table and community feedback.

10. State the core learning of the course.

Technical education becomes socially useful when students understand local knowledge respectfully, analyse it systematically and develop improvements with the community.

Local knowledge ആദരവോടെ പഠിച്ച് community-യുമായി ചേർന്ന് practical improvement നൽകുമ്പോഴാണ് technical education സാമൂഹികമായി പ്രയോജനപ്പെടുന്നത്.

REFERENCES FROM THE SYLLABUS



# Text and reference material

Course 4001

| Cod<br>e | Book / Source                                                                                                                                                  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1       | Nirmal Sengupta, <i>Traditional Knowledge in Modern India</i> , Springer.                                                                                      |
| R2       | Kerala State Planning Board, <i>Economic Review</i> , relevant year.                                                                                           |
| R3       | M. Lalithambika, <i>Techno-socio-economic survey on the living and working conditions of the traditional potter communities of Kerala</i> , IRTC Publications. |
| R4       | A. Sreedhara Menon, <i>Legacy of Kerala</i> , DC Books.                                                                                                        |